

Lecture 10:

Working in the Enterprise: Systems, Certifications, and Careers

Information Technology Fundamentals
School of Information Technology
King Mongkut's Institute of Technology Ladkrabang

เก่งมากถ้าสามารถอ่านหนังสือได้

Objectives Overview

ข้อได้กษณ

บอกคุณน้ คอย
ร่กในนลองมากกับ

Identify the qualities of
valuable information

Describe various
information systems
used in an enterprise

Identify the
components of and
steps in information
literacy

บอกคุณน้ทำตัวล้นท้น น้คอบ ลูกธักรัณ ธิ บอกคุณน้แม่(จ.บ.)มากกับบ

สัอตัว
แก้ว
มีนเพีย

Describe career
opportunities available
in various segments of
the computer industry

Identify job titles and
responsibilities for
various technology jobs

ขอได้กษณร่ก ถ้ากทุกกับ

ขอได้กษณร่ก (เล็กก่ 44)

ครีๆ ♥

บ๊อเสี่ยของเจ้งของ
ห้ส้งไม่ด
ขอบว้นให้เพื้อ
ประล้าทนอก
ขอบด้า พุดอ๊วไม่เพ
สัอกั ไอฮิงนาฯ คว

สัอจกั

Objectives Overview

Identify mobile
app development
strategies

Identify ways to
prepare for
certification

Describe the
general areas of IT
certification

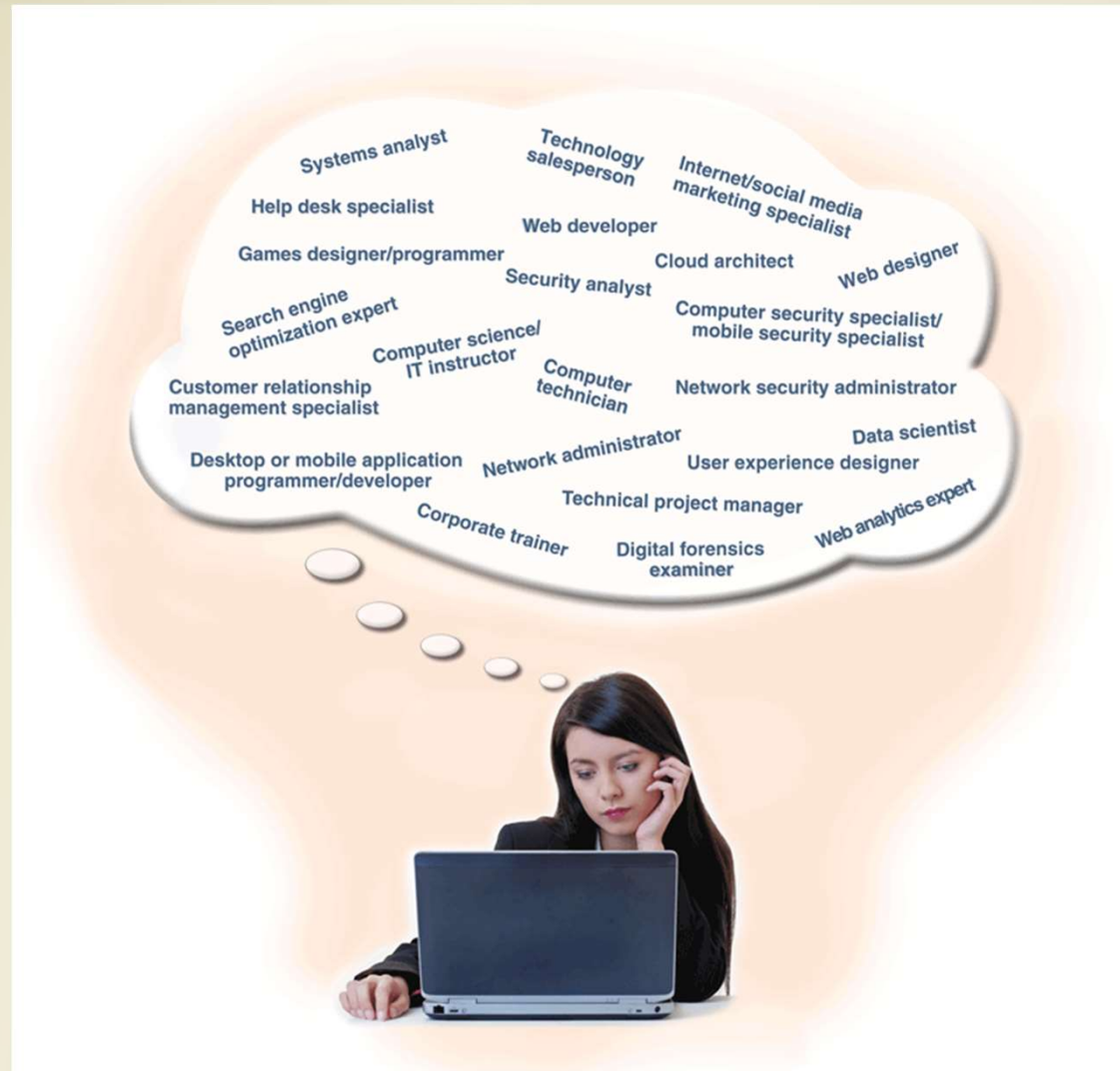
Identify ways to
begin a job search

Explain how to
create a
professional online
presence

The Technology Industry

- Nearly every job requires you to interact with technology to:
 - Complete projects
 - Exchange information with coworkers
 - Meet customers' needs
- As new technologies emerge, organizations look for potential employees who possess skills and a desire to learn and who are comfortable using all types of technology

The Technology Industry



Information Systems in the Enterprise

- An **information system** is a set of hardware, software, data, people, and procedures that work together to produce information
- To ^{ตัวช่วย}assist with sound decision making, information must have value

Information Systems in the Enterprise

: เงินอะไรชอบหาเรื่อง
: เงินจอน!

- For information to be valuable, it should be:

① ค.ถูกต้อง
Accurate
สำคัญ

② ตรวจสอบได้
Verifiable

ค.เป็นปจพ.
Timely
ข้อมูลเราต้องจปจพ.

ก.จัดกลุ่ม
Organized

- categorize
- sort
- filter ก.กรอ (ที่เราสนใจ)
- format (column, row, ...) ทำreport

เข้าถึงง่าย
Accessible

มีปจ.
Useful

ต้นทุนข้อมูลที่ได้มา
Cost effective
ข้อมูลสำคัญ) ที่ไม่ถูกค่า

- relevant มีค.เกี่ยวข้อง
- useful for someone, some situations/cases

Information Systems in the Enterprise

แผนก / ภาควิชา / กรม / กอง

- A typical enterprise consists of a wide variety of departments, centers, and divisions — collectively known as functional units



พนักงาน / บุร

Information Systems in the Enterprise

ຊຳລັບລະບົບ = ສໍາເລັດໄດ້ ທຳກິດໄດ້

- **Enterprise Resource Planning (ERP)** integrates MRP II with the information flow across an organization to manage and coordinate the ongoing activities of the enterprise ທ່າວຮ່ວມກະໂຈນສໍາເລັດໄດ້



Information Systems in the Enterprise

BALLON

NA
HEE

เด็กน้อย



ไม่สนใจมาก
บ้างก็ > <
สนใจบ้าง

Information Systems in the Enterprise

Document Management System (DMS)

- Allows for storage and management of a company's documents
- Supports access control, security, version tracking, and search

Content Management System (CMS)

- Enables and manages the publishing, modification, organization, and access of various forms of documents and other files
- Provides security controls
- Content added through a graphical user interface or webpage

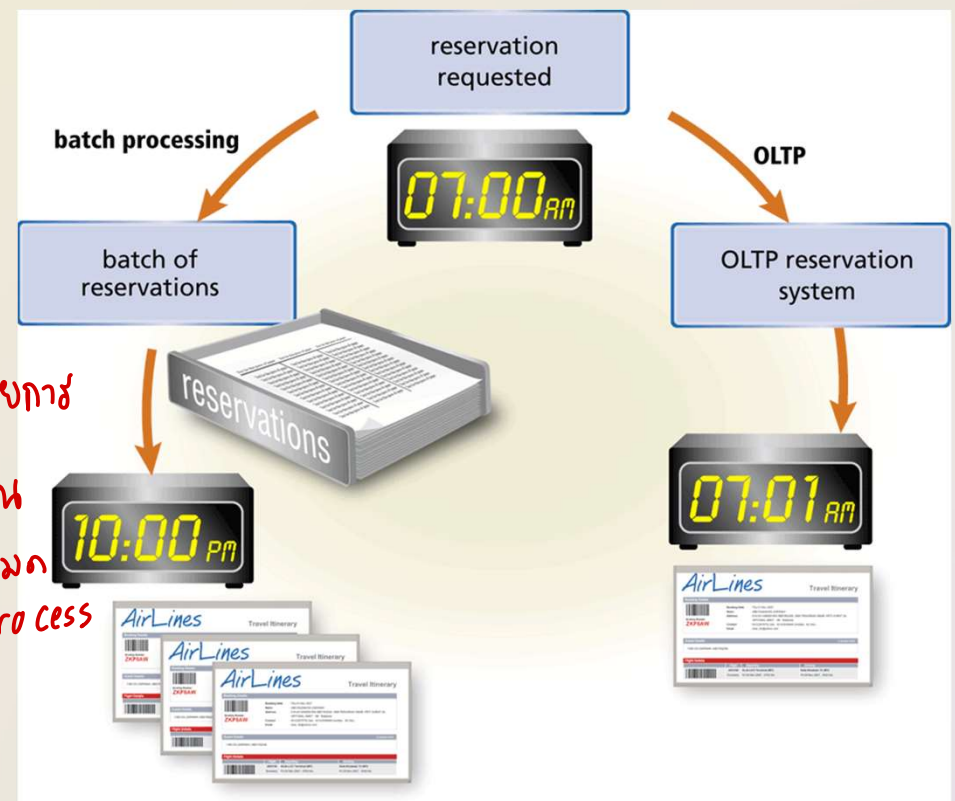
↑ สร้าง web , คอมพิวเตอร์

Information Systems in the Enterprise

- A transaction รายการย่อย processing system (TPS) is an information system that captures and processes data from day-to-day business activities
 - Online transaction processing (OLTP)

- พกชื่อนั้น

- เก็บทุกๆ รายการ
- เก็บเป็นก้อน
รอบรวมมาทั้งหมด
ค่อยมา process



Information Systems in the Enterprise

ทำรายงาน report

- A **management information system (MIS)** is an Information system that generates accurate, timely, and organized information, so that managers and other users can make decisions, solve problems, supervise activities, and track progress

สรุป

Information Systems in the Enterprise

Detailed Flight Report for Flight #328				
Passenger Name	Gender	Birthdate	Seat	Premier Club
Adams, Latisha	F	4/25/92	3C	Y
Brewer, Milton	M	10/14/45	22F	N
Cam, Lin	F	12/16/91	2A	Y
Canaan, Lana	F	4/12/90	21A	N
Cole, Kristina	F	5/10/79	16C	N
Drake, Louella	F	3/4/81	4A	Y
Galens, Lynette	F	11/2/75	2C	N
Gilbert, Laura	F	2/20/78	4F	N
Henreich, Max	M	3/10/85	17C	Y
Hidalgo, Ronald	M	10/15/44	3F	Y
Marsh, Constance	F	11/5/82	2C	N
McGill, Teresa	F	2/27/73	16F	Y
Moretti, Leo	M	9/22/90	17A	Y
Nitz, Dawn	F	7/12/65	3F	N
Ruiz, Albert	M	2/13/93	10D	Y
Stein, Michelle	F	8/16/50	3A	N
Tu, Benjamin	M	1/16/77	22C	N
Van Wijk, Fred	M	6/9/89	10A	Y
Warner, Betty	F	7/1/58	16A	N

101 TPS มาส่ง

Summary Flight Report for March 30			
Flight #	Origin/ Destination	Passengers	Premier Club Members
1048	ORD – RSW	108	33
543	ORD – BMI	24	12
715	ORD – LAX	160	62
701	ORD – JFK	26	10

report ที่ detail ของ member

Exception Flight Report for March 30				
Flight #	Class	Origin/ Destination	Premier Club Members	Premier Club Member Goal
1048	A	ORD – RSW	1	4
701	C	ORD – JFK	3	5

Information Systems in the Enterprise

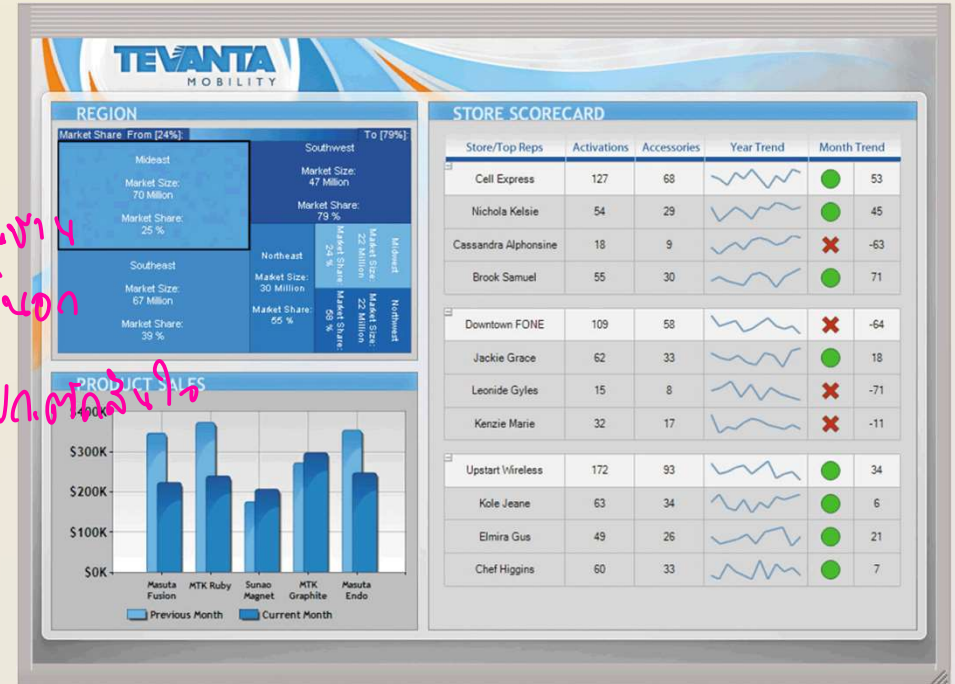
↑ อัตราผลตอบแทน/ส่วน/ เงินทุน (ราคาหลักทรัพย์)
ระบบสารสนเทศ

- A **decision support system (DSS)** helps users **analyze** information and make decisions

- Online analytical processing (**OLAP**)

เอาข้อมูลพื้นฐาน
ไปวิเคราะห์

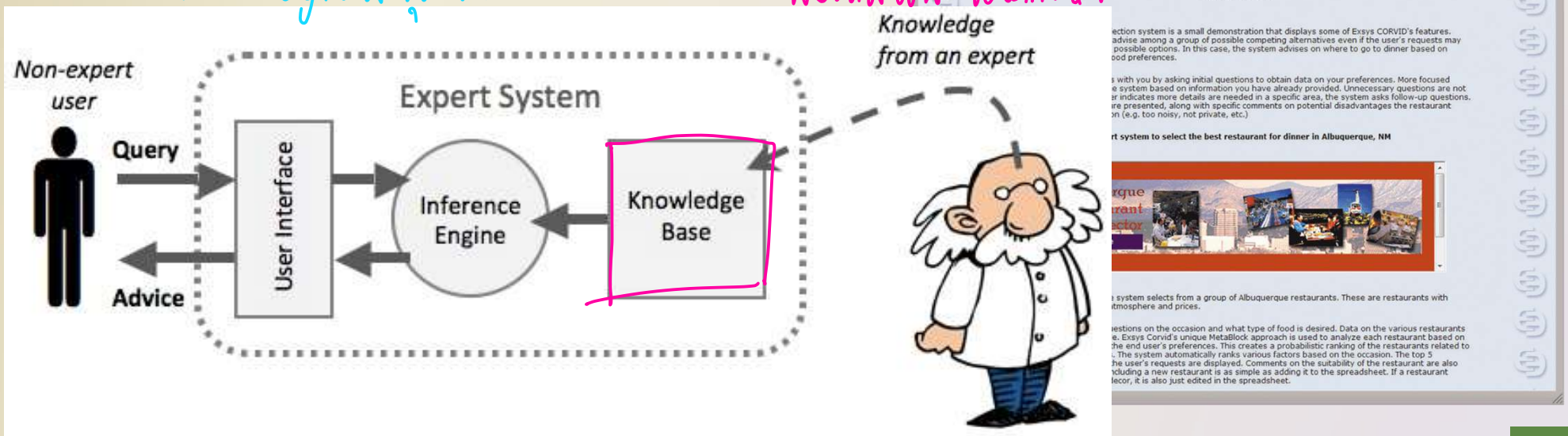
ปร: กอช. ตัดสินใจ



- A DSS uses data from internal and external sources.
- Data from external sources could include interest rates, population trends, or raw material pricing.

Information Systems in the Enterprise

- An **expert system** is an information system that captures and stores the knowledge of human experts and then imitates human reasoning and decision making



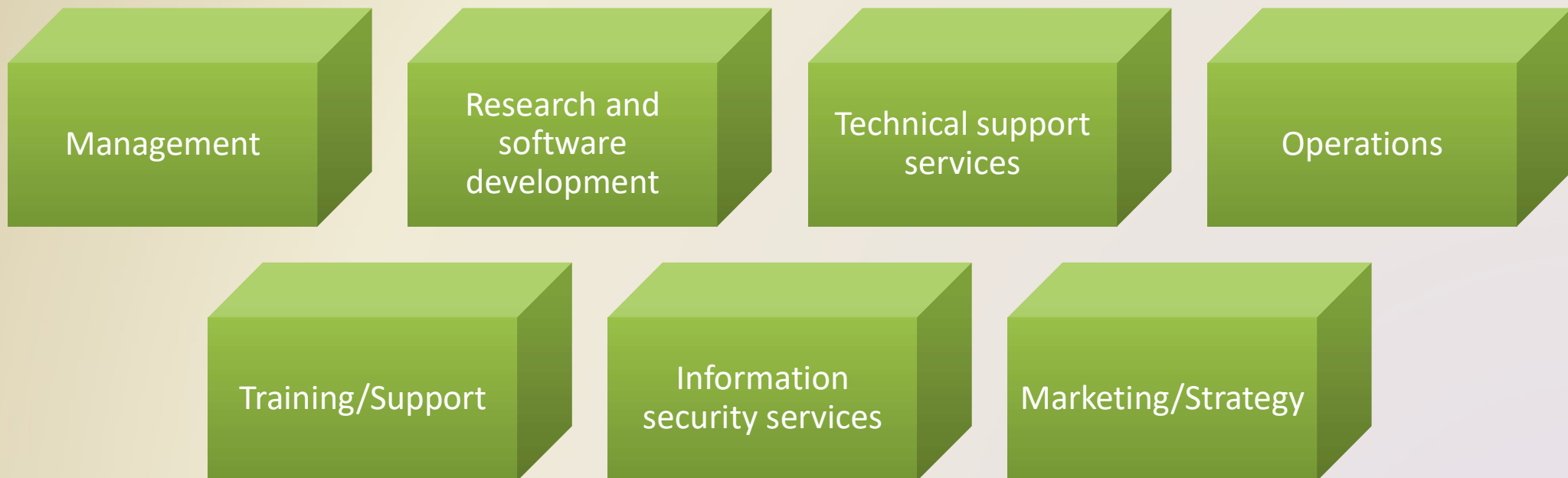
Technology Careers

- Business and government organizations of all sizes use a variety of computers, mobile devices, and other technology
- Larger organizations use computers and other technology to answer and route phone calls, process orders, update inventory, and manage accounts receivable, accounts payable, billing, and payroll activities

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Technology Careers

- IT jobs usually are divided into the following areas:



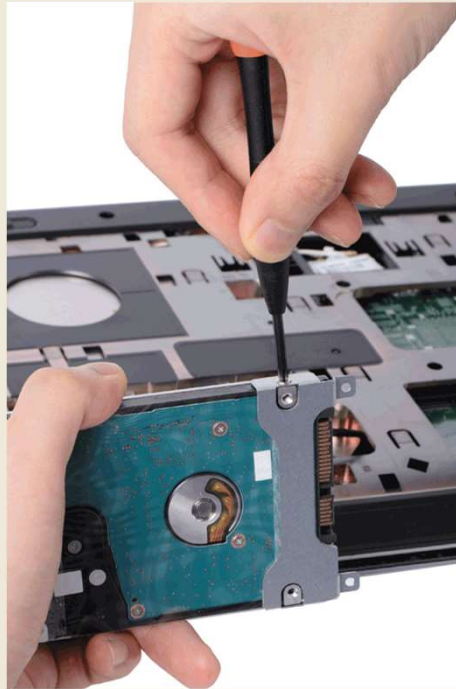
Technology Careers

- The technology equipment field consists of manufacturers and distributors of computers, mobile devices, and other hardware, such as magnetic and optical drives, monitors, printers, and communications and networking devices
- The software and apps field consists of companies that develop, manufacture, and support a wide range of software and apps for computers, the web, and mobile devices



Technology Careers

- The technology service and repair field provides preventive maintenance, component installation, and repair services to customers



Technology Careers

- Technology salespeople must possess a general understanding of technology and a specific knowledge of the product they are selling



Technology Careers

- Schools, colleges, universities, and private companies all need qualified educators to provide technology-related education and training
- Corporate **trainers** teach employees how to use **software** and **apps**, **design** and **develop** systems, write programs, integrate and sync data from apps used on multiple devices, and perform other technology-related activities



Technology Careers

- A help desk specialist answers hardware, software, and networking questions in person, over the phone, or electronically via email or a chat room
- An IT consultant, typically hired based on expertise, provides technology services to his or her clients

ถาม - ตอบ เกี่ยวกับ

Technology Careers

Table 12-2 System Development Jobs

Job Title	Job Description
Cloud Architect ออกแบบ	Identifies business requirements, strategies, and solutions for cloud storage and services that meet a company's goals or needs
Database Designer	Specifies the structure, interface, and requirements of a large-scale database; determines security and permissions for users
Program and App Developer	Specifies, designs, implements, tests, and documents programs and apps in a variety of fields, including robotics, operating systems, animation, and applications
Systems Analyst วิเคราะห์	Works closely with users to analyze their requirements, designs and develops new information systems, and incorporates new technologies
Systems Programmer	Installs and maintains operating system software and provides technical support to the programming staff
Web Designer	Designs the layout, navigation, and overall appearance of a website with a focus on user experience; specifies a website's appearance using HTML5, JavaScript, CSS, media, and other web design technologies
Web Developer	Analyzes, develops, and supports the functionality of a website, including applications that often interact with databases or other online resources

จัดหมวดหมู่:
อาชีพงานด้านไอที

Technology Careers

Table 12-3 Technology Operations Jobs

Job Title	Job Description
Computer Technician	Installs, maintains, and repairs hardware and servers; installs, upgrades, and configures software; troubleshoots hardware problems
Help Desk Specialist/ Help Desk Technician	Answers technology-related questions in person, on the phone, or via email or an online chat room
Network Administrator/ Engineer	Installs, configures, and maintains LANs, WANs, wireless networks, intranets, Internet systems, and network software; identifies and resolves connectivity issues
Technical Project Manager	Guides design, development, and maintenance tasks; serves as interface between programmers/developers and management

Technology Careers

Table 12-4 Web Marketing and Social Media Jobs

Job Title	Job Description
Customer Relationship Management (CRM) Specialist	Integrates apps and data related to customer inquiries, purchases, support requests, and behaviors in order to provide a complete application that manages a company's relationships with its customers
Internet/Social Media Marketing Specialist	Directs and implements an organization's use of Internet and social media marketing, including Facebook pages, Twitter feeds, blogs, and online advertisements
Search Engine Optimization (SEO) Expert	Writes and develops web content and website layouts so that they will appear at the beginning of search results when users search for content
User Experience (UX) Designer	Plans and designs software and apps that consider a user's reaction to a program and its interface, including its efficiency, its effectiveness, and its ease of use

↑ ออกแบบ app
โดยคำนึงถึง user

Technology Careers

Table 12-5 Data Storage, Retrieval, and Analysis Jobs

Job Title	Job Description
Data Scientist	Uses analytics and other Big Data tools to compile statistics on data that an organization can use to plan product development or create strategies for marketing
Database Analyst	Uses data modeling techniques and tools to analyze and specify data usage
Database Administrator	Creates and maintains the data dictionary; monitors database performance
Digital Forensics Examiner	Collects and analyzes evidence found on computers, networks, mobile devices, and databases
Web Analytics Expert	Collects and measures Internet data, such as website traffic patterns and advertising, and develops reports that recommend strategies to maximize an organization's web presence

Technology Careers

Table 12-6 Information and Systems Security Jobs

Job Title	Job Description
Computer Security Specialist/ Mobile Security Specialist	Responsible for the security of data and information stored on computers and mobile devices within an organization
Network Security Administrator	Configures routers and firewalls; specifies web protocols and enterprise technologies
Security Analyst	Implements security procedures and methods, looks for flaws in security of a company's devices and networks, works with and trains employees at all levels, and assigns permissions and network settings
Security System Project Manager	Develops and maintains programs and tools designed to provide security to a network
Digital Forensics Analyst	Inspects electronic data to recover documents and files from data storage devices that may have been damaged or deleted, in order to use them as evidence in a crime investigation

Technology Careers

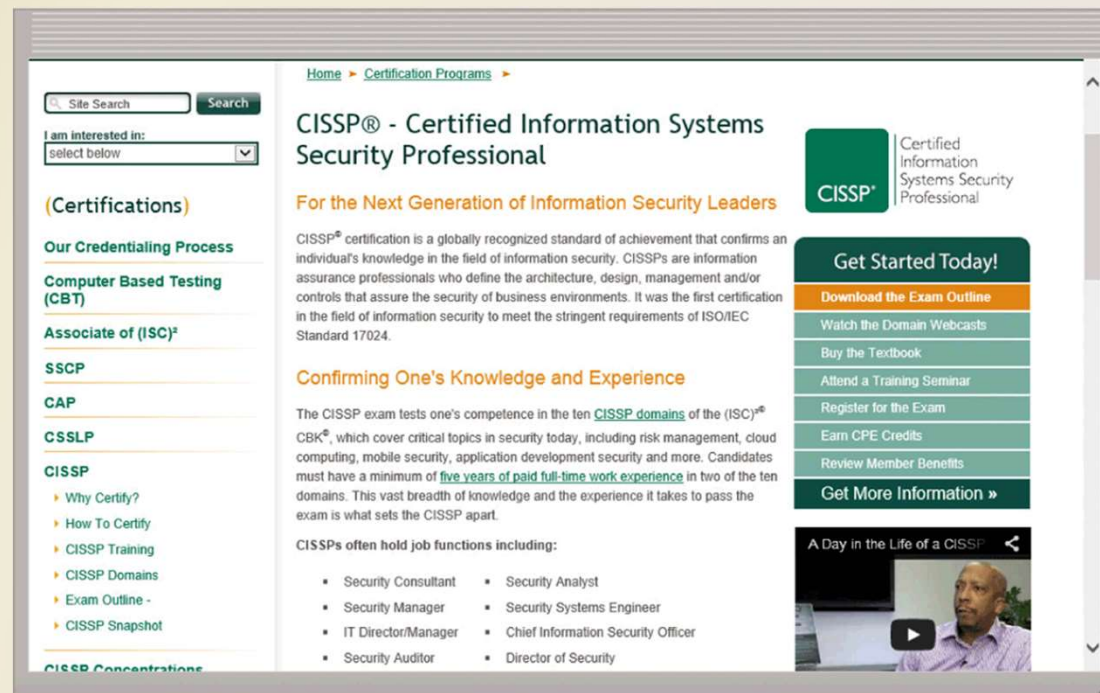
พัฒนา app / ออกแบบ Game

Table 12-7 App Development and Mobile Technologies Jobs

Job Title	Job Description
Desktop or Mobile Application Programmer/Developer	Converts the system design into the appropriate application development language, such as Visual Basic, Java, C#, and Objective C, and toolkits for various platforms
Games Designer/Programmer	Designs games and translates designs into a program or app using an appropriate application development language
Mobile Strategist	Integrates and expands the company's initiatives for mobile users
Mobile Technology Expert	Develops and directs an organization's mobile strategy, including marketing and app development

Technology Certifications

- A certification demonstrates your knowledge in a specific area to employers or potential employers
- Obtaining a certification requires time and money



Technology Certifications

- Application Software Certifications

Corporate trainers

Help desk
specialists

Office
managers/workers

Technology sales
representatives

Technology
teachers

Technology Certifications

- Data Analysis and Database Certifications

Data scientist

Database
administrators

Database
analysts

Database administrators
Digital
forensics
examiners

Technology Certifications

- Hardware Certifications

Cable
installation
technicians

Computer
repair
technicians

Corporate
trainers

Help desk
specialists

IT consultants

System
engineers and
administrators

Technology Certifications

- Networking Certifications

Hardware
service
technicians

IT consultants

Network
managers

Network
engineers

System
administrators

Technology Certifications

WTF

- Operating System Certifications

Hardware
technicians

Help desk
specialists

Network
administrators

IT consultants

System
administrators

Technology Certifications

- Program/Developer Certifications

Game developers

IT consultants

Mobile
application
developers

Project
leaders/managers

Systems analyst

Web developers

Technology Certifications

- Security Certifications

ด้านความปลอดภัย
(เฉพาะด้านที่เฉพาะ)

ด้านความปลอดภัยอื่น ๆ !

Information
security officers
and managers

Law
enforcement
officials

Military
intelligence
officers

Network
administrators

Wireless
network
administrators

Network
security
specialists

Security
administrators

ด้านความปลอดภัย

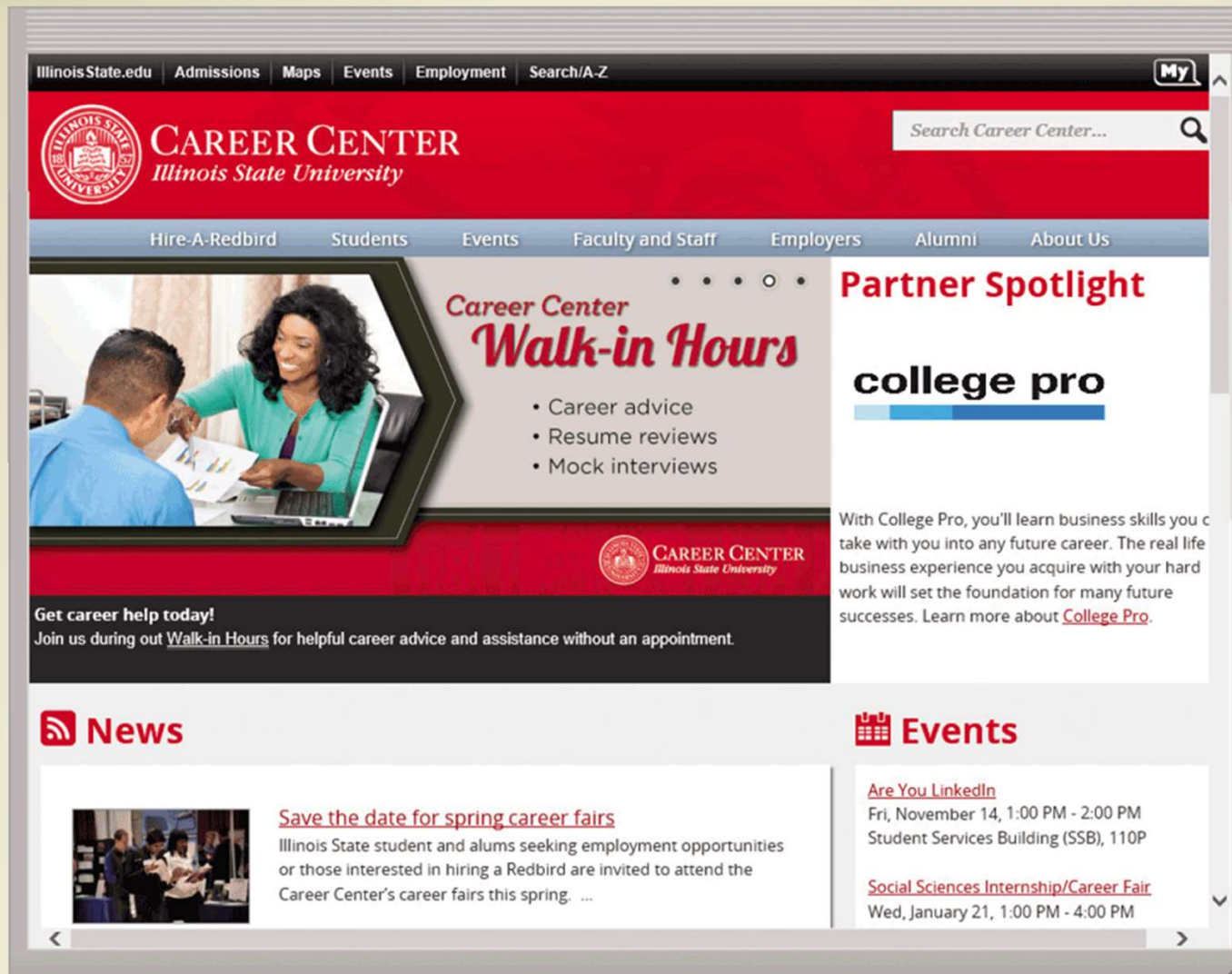
Job Searching and Career Planning

- Many job opportunities may exist in your industry, so it is important to narrow down the available jobs to ones for which you are qualified and in which you are interested
- Tools at your disposal include the career service department at your school, career planning websites, and online social networks
- Whether you are seeking a new job or currently are employed, you may find a career planning website useful

FUCK YOU



Job Searching and Career Planning



Creating a Professional Online Presence

- An understanding of the web, digital media, and online social networks, such as LinkedIn, is beneficial in creating your online presence.
- LinkedIn is an online social network where professionals, such as Reid Hoffman, founder of LinkedIn, can create profiles and connect with coworkers and industry colleagues.
- LinkedIn uses the term, contacts, to describe the individuals in your professional network and also stores your relationship with each contact in your network.

Summary

Information
systems in an
enterprise

Various technology
career fields and
specific technology
jobs

Technology
certifications

How to begin a job
search and create a
professional online
presence

Top 15 IT Certifications in Demand for 2022

ค่าตอบแทน นวัตกรรม

0. กะขี้ ← มาแรงทุกปี อีสาน

1. Certified Data Professional (CDP)
2. Certified Information Systems Security Professional (CISSP)
3. Cisco Certified Internetwork Expert (CCIE)
4. Cisco Certified Network Associate (CCNA)
5. Cisco Certified Network Professional (CCNP)
6. Certified Information Systems Auditor (CISA) – ISACA
7. CompTIA A+
8. Microsoft Technology Associate (MTA)
9. Project Management Professional (PMP) – PMI
10. Oracle Certified Professional
11. Salesforce Development Lifecycle and Deployment
12. Certified Scrum Master (CSM)
13. AWS Certified Solutions Architect
14. Certified Ethical Hacker (CEH)
15. Global Information Assurance Certification (GIAC)

Certified Data Professional (CDP)

- Data Professionals may specialize and certify in many of the following areas. Choose the specialization suited to your experience and career goals;
- The following Job Roles and specialty credentials are catered for within this credential
 - Business Analytics, (CDP-BA)
 - Data Analytics & Design, (CDP-DAD)
 - Data Governance, (CDP-DG; DGSP)
 - Data Integration and Interoperability), (CDP-DII)
 - Data Management, (CDP-DM)
 - Data & Information Quality, (CDP-DIQ)
 - Data Stewardship, (CDP-DS)
 - Data Warehousing, (CDP-DW)
 - Enterprise Data Architecture (Zachman), (CDP-EDA)
 - Information Systems or IT Management, (CDP-Mgmt)
 - Public Sector Data Governance (CDP-PSDG; PSDGP)

Cisco Certified Network Associate (CCNA)

- CCNA exam covers networking fundamentals, IP services, security fundamentals, automation and programmability.
- Designed for agility and versatility, CCNA validates that you have the skills required to manage and optimize today's most advanced networks.

CompTIA A+

- CompTIA A+ certified professionals are proven problem solvers. They support today's core technologies from security to networking to virtualization and more. CompTIA A+ is the industry standard for launching IT careers into today's digital world.
- CompTIA A+ is the only industry recognized credential with performance testing to prove pros can think on their feet to perform critical IT support tasks. It is trusted by employers around the world to identify the go-to person in end point management & technical support roles. CompTIA A+ appears in more tech support job listings than any other IT credential.

CISSP

- The CISSP is ideal for experienced security practitioners, managers and executives interested in proving their knowledge across a wide array of security practices and principles, including those in the following positions:
 - Chief Information Security Officer (CISO) *ศาน. สู่สุด แงด ๗*
 - Chief Information Officer (CIO)
 - Director of Security
 - IT Director/Manager
 - Security Systems Engineer
 - Security Analyst
 - Security Manager
 - Security Auditor
 - Security Architect
 - Security Consultant
 - Network Architect

Certified Ethical Hacker (CEH)

- Our exclusive Learn | Certify | Engage | Compete framework covers not only a comprehensive training program to prepare you for the certification exam, but also the industry's most robust, in-depth, hands-on lab experience of any cybersecurity program available.
- C|EH v12 will teach you the latest commercial-grade hacking tools, techniques, and methodologies used by hackers and information security professionals to lawfully hack an organizations.

Cyber Careers

- Cyber Defense Specialist
- Cyber Defense Infrastructure Engineer
- Cyber Security Engineer
- Cyber Security Analyst
- Cyber Security Expert
- Cyber Security Consultant
- Cybersecurity and Privacy Consultant
- Cyber Security Manager
- Cyber Security Officer
- Manager, SOC & Cyber Security
- Cyber Security: Vulnerability Analyst & Pen-Tester
- Threat Intelligence and Vulnerability Analyst (TIVA)



Recommended Qualifications

มีพรสวรรค์ในทางด้านไอที

เรามีชีวิตอยู่เพื่อ
ถ้าคนนั้นเก่ง
ไหน อีกคนเก่ง

- Analytical and problem-solving skills
- Have a passion in cybersecurity, IT Security / willing to learn new things
- Good English communication, including conducting presentations and creating reports
- Excellent report writing skills in both English and Thai
- Project management and communication skill
- To be certified for Network, Infrastructure, Security related certification would be advantageous

ปลอ่งกุสัท

ส่วน IT

มีใครบ้างที่สนใจ

